Axient: Empirical Calibration of Venue-Agnostic Event-Margin Protocols:

A Prospectively Frozen Analysis Plan, Deterministic Internal-Alpha Technical Pilot, and Venue-Emulator Methodology

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Abstract

The first two Axient studies establish a position-level debt-free-finality mechanism and a protocol-level credit and loss-allocation architecture. They do not show that a live venue supplies the execution depth, settlement reliability, close-window predictability, capital elasticity, or liquidation capacity required for deployment. This paper defines the empirical bridge from mechanism design to controlled operation and reports a deterministic Internal-Alpha technical pilot conducted under a prospectively frozen analysis plan. It makes explicit that a pathwise lower proceeds envelope used by a formal debt-clearing theorem and a conformal lower prediction bound used for empirical validation are different mathematical objects: the latter does not silently replace the former. The design distinguishes quoted, matched, settled, and redeemed value; scheduled and actual close; standalone and aggregate exit capacity; ordinary and adversarial liquidity response; Senior LP and Liquidation Backstop Provider supply; market-maker delivery; liquidation coverage; withdrawal dynamics; reserve use; and leverage eligibility. Calibration and evaluation windows are separated, shared liquidity is measured on one aggregate execution curve, and no-go outcomes remain admissible. The pilot contains twelve synthetic emulated-execution journeys: ten were admitted and completed, two were rejected before admission, and all were terminally classified. Expected and observed state hashes matched across the exact model, contracts, services, and API projections; unexplained reconciliation drift, duplicate-event financial delta, conflicting-evidence debt delta, external venue writes, and production asset movement were zero. Browser, accessibility, permission, and fault-recovery suites passed. These results validate deterministic instrumentation and integration for

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registered synthetic paths only. They do not identify live execution or settlement latency, human usability, provider demand, yield, default probability, or production readiness. The paper therefore supplies a frozen empirical protocol, venue-emulator methodology, technical-pilot evidence, and explicit validity gates for future sandbox and live-venue evaluation, while keeping deterministic certification separate from probabilistic validation.

Keywords: event markets; empirical calibration; preregistration; execution quality; settlement evidence; venue emulator; leverage eligibility; internal alpha.
JEL Classification: C12, C14, C18, G13, G14, G23.

1 Introduction

Formal mechanisms can be internally correct and practically unusable. A debt-clearing theorem may hold inside a registered operating set even if a live venue rarely supplies the required depth, settles too slowly, closes earlier than scheduled, or does not provide enforceable signing and collateral controls. The empirical task is therefore not to "validate the theorem" by estimating one average. It is to identify the operating region in which the assumptions are observed, measure how often and how severely they fail, and determine whether any commercially useful leverage tier survives.

The study sits between market microstructure, survival analysis, conformal prediction, capital-supply estimation, and reproducible research design [Glosten and Milgrom, 1985, Kyle, 1985, Hasbrouck, 2007, Cox, 1972, Kaplan and Meier, 1958, Lei et al., 2018, Nosek et al., 2018]. It also responds to a recurrent problem in prediction-market engineering: book values, match records, settlement confirmations, and redeemed cash are different objects, but operational datasets often collapse them into one timestamp or status.

This concise revision has five goals:

(1) define the observation unit, clocks, estimands, and eligibility rules for venue-agnostic event-margin calibration;
(2) lock confirmatory hypotheses, inclusion rules, and validity gates before live-venue analysis;
(3) specify a provenance-aware Venue Emulator methodology for controlled counterfactual and failure testing; and
(4) report a deterministic Internal-Alpha technical pilot without misrepresenting it as live or human evidence; and
(5) separate formal pathwise certification from statistical lower-bound validation so that nominal coverage is never presented as a universal guarantee.

Extended event schemas, estimator registries, power derivations, and blank report shells remain available in the earlier arXiv version history.

2 Empirical object and venue capabilities

The observation unit is

$$z = (v, e, m, a, L, n, t, \pi), \tag{1}$$

where v is venue, e event, m market, a account or policy class, L leverage, n requested size, t decision time, and π the registered execution policy. Repeated observations are clustered by market, event, venue, close window, and shared liquidity source.

Definition 2.1 (Policy operating set and pathwise certificate)**.** *For policy version θ, let $\mathcal{U}_\theta^{\mathrm{op}}$ be the prospectively registered set of paths on which the formal control assumptions are required to hold. For policy Π and quantity x, define the pathwise settled-proceeds envelope*

$$\underline{B}_\theta^{\Pi}(x) = \inf_{\omega \in \mathcal{U}_\theta^{\mathrm{op}}} B^{\mathrm{set},\Pi}(x, \omega). \tag{2}$$

A deterministic debt-clearing certificate compares this envelope with a debt-service upper envelope and a registered buffer on the same path domain.

Definition 2.2 (Empirical validation region)**.** *For confidence level $1-\alpha$, the empirical validation region $\widehat{\mathcal{O}}_{\theta,\alpha}$ is the subset of observations for which capability, data-quality, execution, settlement, control, and sampling gates are observed and a registered statistical lower bound clears debt plus the policy buffer. Membership is an empirical support statement at the declared coverage level; it is not a pathwise guarantee over $\mathcal{U}_\theta^{\mathrm{op}}$.*

A venue capability vector records at least:

$$\chi_v = (\text{book}, \text{write}, \text{custody}, \text{lien}, \text{liquidation}, \text{settlement proof}, \text{close}, \text{redemption}). \tag{3}$$

Missing capabilities cannot be inferred from a public API or a successful read-only connection. Evidence is monotone: stronger verified capability may support stronger claims; weaker evidence cannot be substituted for a missing control.

The principal gates are:

(G1) **BOOK**: reconstructable, timestamped, sequence-consistent execution data;
(G2) **SETTLE**: accounting-grade settlement evidence distinct from match;
(G3) **CLOSE**: observable scheduled and actual close;
(G4) **SIGN**: enforceable risk-reducing signing authority;
(G5) **LIEN**: financed collateral cannot escape while debt is positive;
(G6) **CAPITAL**: Senior and LBP capital are identifiable and segregated;
(G7) **MM**: commitment terms and delivery can be measured;
(G8) **SAMPLE**: effective sample size and cluster coverage satisfy the design.

A venue or market can fail the study without invalidating the underlying mechanism; the empirical conclusion may simply be that the operating region is empty.

3 Data architecture, provenance, and clocks

3.1 Sampling and stratification

Sampling is stratified before outcome analysis by venue, event class, market, time-to-close, spread/depth regime, requested quantity, leverage tier, and capability class. The design stores both the attempted population and every exclusion reason. Markets are not removed merely because they fail to settle, close early, lack a signer boundary, or produce an empty leverage set; those are primary outcomes.

A calibration window is used to fit execution and settlement envelopes, while a disjoint evaluation window estimates coverage and breach severity. Repeated observations from the same market or event are clustered. If the observation process samples order books at high frequency, effective sample size is governed by cluster count and dependence rather than raw rows. Sampling weights, if used, are frozen before evaluation.

The canonical record contains a root decision event and a terminal classification. Every attempted observation must end as admitted/completed, rejected, failed, abandoned, crashed, or unresolved at the administrative cutoff. The equality

$$N_{attempted} = N_{completed} + N_{rejected} + N_{failed} + N_{abandoned} + N_{crashed} + N_{unresolved} \quad (4)$$

is a release gate. This prevents a technical pilot from reporting only successful traces.

The data model separates source payloads, normalized observations, derived features, protocol decisions, and terminal evidence. Every record carries source, retrieval time, schema version, content hash, rights classification, and transformation lineage. Raw third-party payloads are not redistributed unless source-specific rights permit it.

The four value states are:

$$V^{\text{book}} \to V^{\text{match}} \to V^{\text{settled}} \to V^{\text{redeemed}}. \quad (5)$$

No weaker state can substitute for a stronger one in debt-repayment analysis.

Important clocks are:

$$\begin{aligned} t_d &= \text{decision time}, & t_o &= \text{order submission}, & (6)\\ t_m &= \text{match}, & t_s &= \text{settlement confirmation}, & (7)\\ t_c^{\text{sched}} &= \text{scheduled close}, & t_c^{\text{act}} &= \text{actual close}, & (8)\\ t_f &= \text{final payout}, & t_r &= \text{redemption}. & (9) \end{aligned}$$

For Internal Alpha, scenario clocks are synthetic and cannot estimate operational latency.

Execution deterioration ratios are defined only when denominators are positive:

$$\rho_{bm} = \frac{V^{\text{match}}}{V^{\text{book}}}, \quad \rho_{ms} = \frac{V^{\text{settled}}}{V^{\text{match}}}, \quad \rho_{sr} = \frac{V^{\text{redeemed}}}{V^{\text{settled}}}. \quad (10)$$

The study reports the distribution and tails of each stage separately rather than one end-to-end average that hides the failure location.

4 Execution, settlement, and shared liquidity estimands

For requested sale quantity x, let $B_z(x)$ be book-implied net proceeds and $S_z(x)$ eventually settled proceeds. Principal execution estimands include:

- the conditional shortfall $B_z(x) - S_z(x)$;
- the ratio $S_z(x)/B_z(x)$;
- settlement failure and competing-risk probabilities;
- match-to-settlement and match-to-redemption survival curves;
- actual-close lead relative to scheduled close; and
- the probability that a debt-clearing quantity remains executable through the hard-flat horizon.

A split-conformal lower envelope is one candidate calibration method. Fit a predictive score on a calibration window and choose a residual quantile so that

$$\underline{S}_{1-\alpha}(x \mid Z) = \widehat{S}(x \mid Z) - q_{1-\alpha} \tag{11}$$

has marginal coverage on a disjoint evaluation set under exchangeability conditions [Vovk et al., 2005, Lei et al., 2018]. Cluster and time dependence require block or group adaptations; nominal marginal coverage is not automatically conditional coverage at rare close states.

4.1 Deterministic certification versus probabilistic validation

The objects in Equations (2) and (11) have different quantifiers. The pathwise envelope is an infimum over every path in a registered operating set. The conformal quantity is a data-dependent lower prediction bound with a nominal marginal coverage statement under exchangeability.

Proposition 4.1 (A conformal lower bound does not imply a pathwise infimum)**.** *Marginal coverage*

$$\mathbb{P}\{S(x, Z) \geq \underline{S}_{1-\alpha}(x \mid Z)\} \geq 1 - \alpha \tag{12}$$

does not in general imply

$$\underline{S}_{1-\alpha}(x \mid Z) \leq \inf_{\omega \in \mathcal{U}_\theta^{\mathrm{op}}} B^{\mathrm{set},\Pi}(x, \omega). \tag{13}$$

Proof. Let an admissible operating set contain a rare path $\omega^\star$ with proceeds below the conformal bound, while the distribution used for marginal coverage assigns probability at most α to that path. Then Equation (12) may hold although the infimum over the operating set is attained at $\omega^\star$ and violates Equation (13). Marginal probability and universal path quantification are therefore not interchangeable. □

Table 1: Certification and validation claims are not substitutes.

Object	Guarantee	Permitted use
$\underline{B}_\theta^\Pi$	Pathwise over registered $\mathcal{U}_\theta^{\mathrm{op}}$	Formal certificate under the mechanism assumptions
$\underline{S}_{1-\alpha}$	Marginal statistical coverage under declared exchangeability	Held-out validation, candidate probabilistic admission, breach measurement
Observed settled proceeds	Exact for the observed evidence record	Ex-post reconciliation and empirical endpoint

The registered design uses the two objects as follows. Formal debt-free-finality results retain the deterministic envelope $\underline{B}_\theta^\Pi$. Empirical data test the frequency and severity with which the policy envelope or a separately labelled candidate statistical envelope is breached. A conformal envelope may support a probabilistic admission mode only when the residual α-risk is named explicitly and propagated into capital, reserve, and loss-allocation policy. The present technical pilot does not make that product claim.

Multiple positions share one execution curve. If x_i are contemporaneous sale quantities and $B(x)$ is concave or stepwise, generally

$$B\left(\sum_i x_i\right) \leq \sum_i B_i(x_i), \tag{14}$$

where the right-hand side re-anchors the same book for each position.

Proposition 4.2 (Shared-book non-additivity)**.** *When at least two positions consume the same favorable depth and their combined order reaches a worse level, independent position valuation strictly overstates aggregate settled proceeds.*

5 Capital supply and leverage eligibility

Senior LP supply and LBP supply are estimated separately because they bear different priority, lock-up, and impairment. A generic semi-elasticity specification is

$$\log S_{p,t} = \alpha_p + \beta_r r_{p,t} - \beta_\ell \widehat{\ell}_{p,t} - \beta_q Q_{p,t} - \beta_o r_t^{\mathrm{out}} + u_{p,t}, \tag{15}$$

where Q is queue friction and r^{out} an outside opportunity. Identification requires exogenous rate or policy variation; observational associations are not automatically supply curves.

Market-maker commitments are evaluated by promised and delivered quantity, price, validity interval, concentration, bond quality, and stress-state delivery. Liquidator

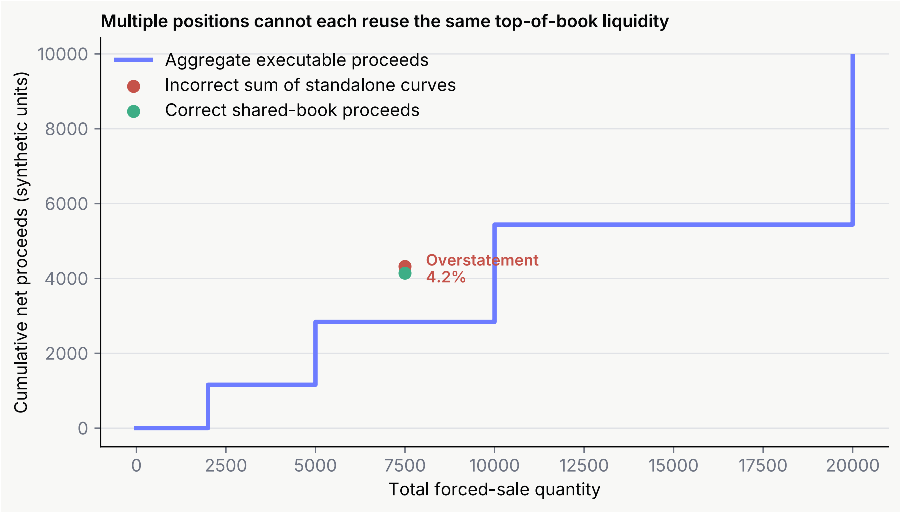

Figure 1: Standalone estimates reuse favorable depth; aggregate execution consumes the shared curve once.

coverage is the fraction of required risk-reducing quantity for which an eligible executor is available at the registered bounty.

For leverage L, the linear debt-recovery benchmark is

$$g_{\min}(L) = 1 - \frac{1}{L}. \tag{16}$$

Thus $2\times$, $3\times$, and $5\times$ require at least 50%, 66.7%, and 80% net recovery before additional fees and buffers. Exact eligibility uses discrete entry and aggregate exit curves, not this scalar benchmark.

Definition 5.1 (Certified and empirically supported leverage)**.** *A tier L is* formally certified *for observation class z only if all capability and mechanism gates pass and the deterministic envelope $\underline{B}_\theta^\Pi$ covers debt plus the registered buffer on $\mathcal{U}_\theta^{\mathrm{op}}$. A tier is* empirically supported at level $1-\alpha$ *when the corresponding held-out validation gates, including coverage and breach-severity requirements, pass. The second label does not upgrade the first.*

The effective tier is the maximum tier satisfying the registered decision rule in a fixed policy set, with the claim type (formal certificate or probabilistic support) reported explicitly and no extrapolation. An empty set is a valid result. During early operation, $3\times$ and $5\times$ may be computed as shadow-only outcomes while execution remains capped at $2\times$.

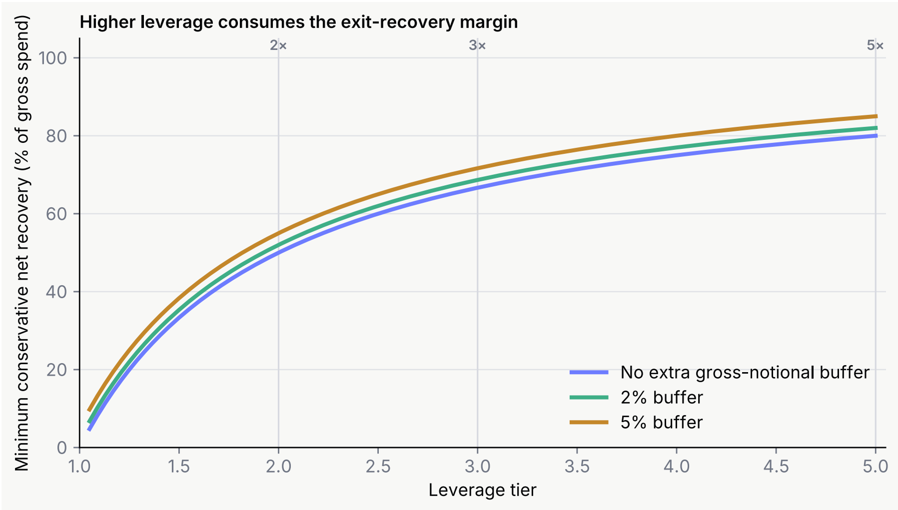

Figure 2: Linear recovery benchmark by leverage. Actual eligibility is book-dependent and includes additional buffers.

6 Frozen hypotheses, validity gates, and sample design

The analysis plan was frozen before detailed trace-level and aggregate scientific outputs were interpreted. The public registration status should therefore be described as a prospectively frozen internal analysis plan unless and until an external registration is completed.

Confirmatory hypotheses cover:

(H1) held-out coverage and breach severity for the registered proceeds rule, without treating nominal coverage as pathwise certification;
(H2) settlement reliability within registered horizons;
(H3) hard-flat completion before actual close;
(H4) explanatory value of actual-close indicators beyond scheduled close;
(H5) aggregate versus standalone liquidity divergence;
(H6) non-empty 2× eligibility under approved capabilities;
(H7) stricter evidence requirements for 3× and 5×;
(H8) market-maker commitment delivery;
(H9) sufficient liquidator coverage;
(H10) request-time neutrality of loss-participating withdrawal queues;
(H11) absence of direct cross-pool ledger contagion; and
(H12) joint viability of provider participation and a restricted 2× operating region.

No result is promoted from exploratory to confirmatory after inspection. Missing mandatory evidence is a failure of the corresponding gate, not a pass obtained by imputation.

Table 2: Compact hypothesis registry.

ID	Object	Confirmatory criterion
H1	Proceeds-rule validation	Held-out coverage meets the registered level, breach severity is reported, and the claim is labelled probabilistic unless a separate pathwise certificate is available.
H2	Settlement reliability	Accounting-grade settlement completes within the registered horizon at the required rate.
H3	Hard-flat	Debt-clearing execution completes before actual close on eligible observations.
H4	Close predictability	Actual-close indicators improve prediction beyond scheduled close alone.
H5	Shared liquidity	Aggregate proceeds are no greater than independently re-anchored proceeds at common size.
H6	2× eligibility	A non-empty eligible set exists under approved capabilities and capital limits.
H7	Higher tiers	3×/5× require strictly stronger evidence and may remain empty.
H8	MM commitments	Delivered stressed capacity satisfies the pre-registered fraction of commitment.
H9	Liquidators	Eligible executor capacity covers the registered liquidation quantity.
H10	Withdrawals	Request timing does not avoid loss recognized before service.
H11	Isolation	No direct local liability transfer occurs across isolated pools.
H12	Viability	Provider participation and restricted 2× demand coexist after costs and losses.

Design-stage calculations provide planning floors. With zero observed failures, the one-sided 95% upper bound falls below 1% after 299 independent observations and below 0.5% after 598. A Dvoretzky–Kiefer–Wolfowitz uniform CDF error of ±5 percentage points at 95% confidence requires 738 independent observations [Dvoretzky et al., 1956, Massart, 1990]. A paired standardized effect of 0.20 requires approximately 199 pairs for 80% power and 265 for 90%, before cluster inflation.

7 Statistical estimators and uncertainty reporting

The empirical programme uses estimators matched to the event process rather than one universal regression. Execution ratios and shortfalls are summarized by medians, quantiles, empirical CDFs, and cluster-aware bootstrap intervals. Settlement and redemption times are analyzed as time-to-event outcomes; unresolved, failed, and competing terminal states are retained rather than discarded.

For event time T and censoring indicator δ, Kaplan–Meier curves describe ordinary completion when censoring assumptions are defensible [Kaplan and Meier, 1958]. When settlement failure, market void, or permanent suspension competes with successful settlement, cumulative-incidence or Fine–Gray-style analyses are reported separately [Fine and Gray, 1999]. Cox-type models may estimate relative hazard associations, but proportional-hazard diagnostics and cluster-robust uncertainty are mandatory [Cox, 1972, Cameron and Miller, 2015].

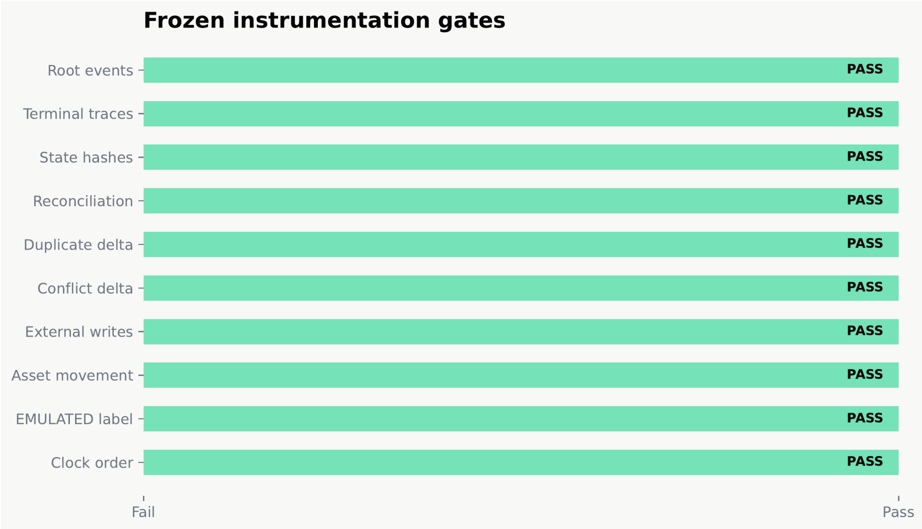

Figure 3: Technical validity gates for the Internal Alpha evidence bundle.

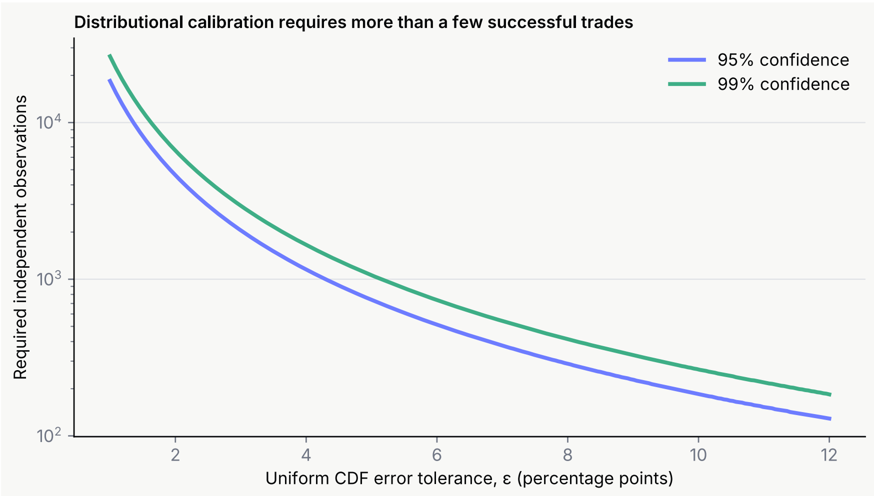

Figure 4: Uniform CDF accuracy under the DKW bound. Cluster dependence increases the required raw observation count.

Coverage analysis reports both nominal success and breach magnitude. If $\underline{S}(z)$ is the registered lower settled-proceeds bound, define

$$B_z = \mathbf{1}\{S_z < \underline{S}(z)\}, \qquad M_z = [\underline{S}(z) - S_z]^+ . \tag{17}$$

A model with few but catastrophic breaches can be less useful than one with more frequent small breaches; both incidence and severity are therefore required.

Capital-supply and market-maker analyses distinguish descriptive response from causal elasticity. Where no valid instrument or randomized policy perturbation exists, the paper reports conditional associations and stress counterfactuals rather than causal supply curves. Multiple-hypothesis adjustment follows a pre-declared family structure, for example Benjamini–Hochberg control for exploratory families [Benjamini and Hochberg, 1995].

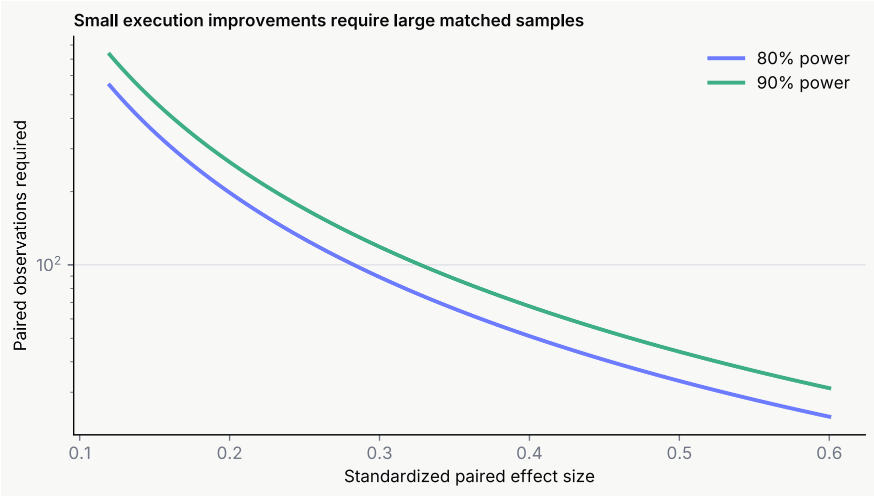

Figure 5: Illustrative paired-design power for standardized effects. These are design-stage calculations, not venue estimates.

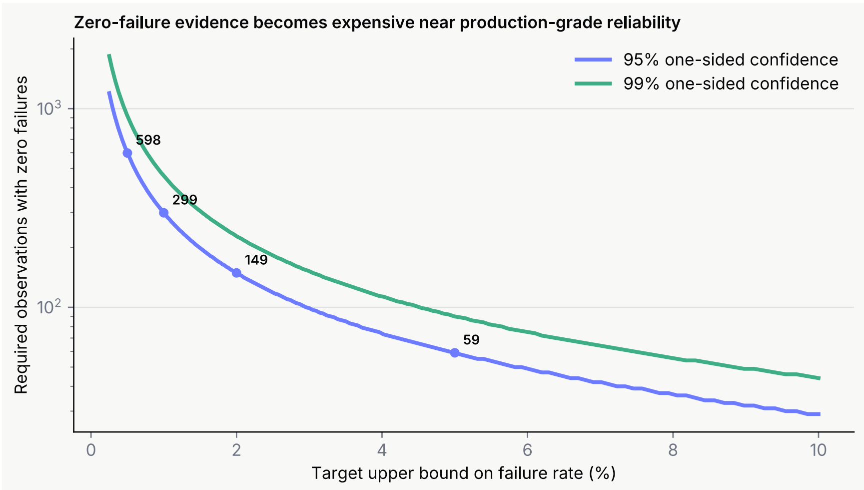

Figure 6: Zero-failure sample size required for selected one-sided upper bounds. Independence assumptions must be adjusted for clustering.

All intervals state their unit of resampling. Market-level and event-level clustering are primary; row-level confidence intervals from high-frequency snapshots are not treated as independent evidence. Sensitivity analysis varies close windows, book aggregation, settlement horizons, and missing-data policies without redefining the main estimand.

8 Venue Emulator methodology

The Event Venue Emulator is a reusable test product that represents venue behavior without containing Axient credit logic. It ingests synthetic or permitted read-only snapshots and emits deterministic market, order, match, settlement, close, resolution,

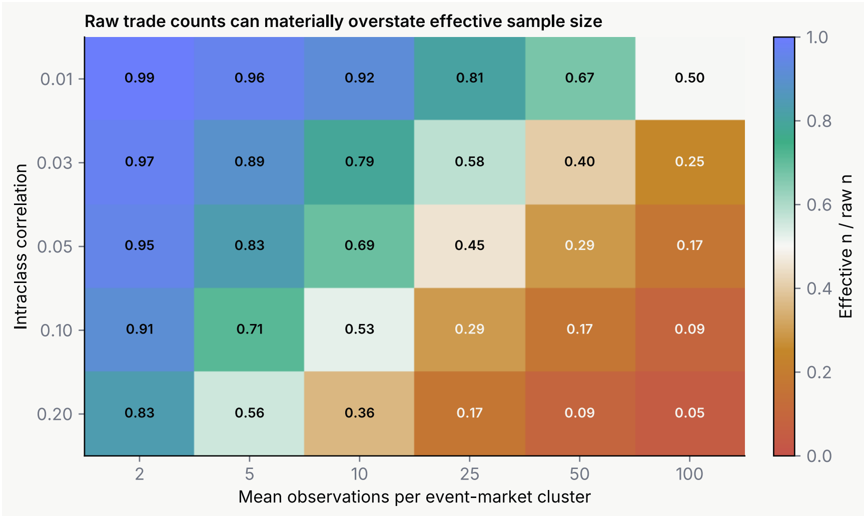

Figure 7: Illustrative design effect from within-cluster dependence. Raw snapshot counts can greatly overstate effective sample size.

redemption, and fault events. Each trace records:

$$\begin{aligned}&(\text{source hash},\ \text{emulator version},\ \text{scenario ID},\ \text{seed},\\ &\ \text{ordered events},\ \text{terminal commitment}).\end{aligned} \tag{18}$$

Proposition 8.1 (Replay identity)**.** *Given the same versioned emulator, source snapshot, scenario definition, and seed, canonical serialization produces the same ordered event stream and trace digest. A different digest is evidence of changed input, implementation, or serialization, not ordinary randomness.*

Emulator outputs are not venue observations. The data model marks 'execution-Mode=EMULATED', records external write count, and separates synthetic, rights-cleared derived, raw-permitted, hash-only, and private-not-distributed provenance classes. The public AVET dataset supplies a canonical trace format and source-anchored synthetic corpus; AEMB supplies compact cross-language conformance scenarios.

9 Internal Alpha technical pilot

The pilot used deterministic local execution, test assets, and a maximum of 2×. It tested instrumentation and cross-layer consistency rather than economic performance. The population accounting is:

The two rejected journeys are part of the denominator and demonstrate fail-closed

Table 3: Internal Alpha journey accounting.

Status	Count
Attempted	12
Admitted	10
Completed	10
Rejected before admission	2
Failed after admission	0
Abandoned	0
Crashed	0
Terminally classified	12

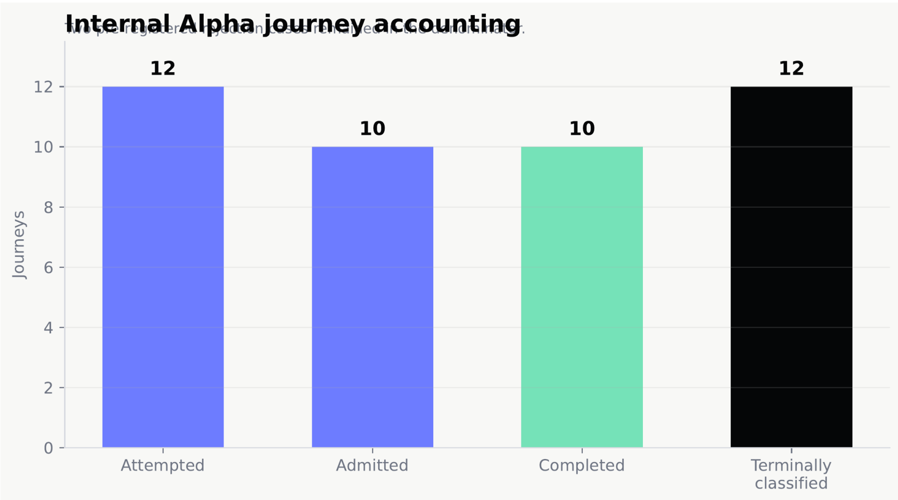

Figure 8: Attempted, admitted, completed, and rejected Internal Alpha journeys. Rejections remain in the denominator.

risk or stale-data controls. The registered technical gates all passed:

$$\begin{aligned} &\text{valid root events} = 100\%, \qquad \text{complete terminal traces} = 100\%, \\ &\text{state-hash equality} = 100\%, \qquad \text{unexplained drift} = 0, \\ &\text{duplicate-event financial delta} = 0, \qquad \text{conflict debt delta} = 0, \\ &\text{external writes} = 0, \qquad \text{production asset movement} = 0. \end{aligned} \tag{19}$$

Nineteen browser journeys, nineteen automated accessibility checks, three permission tests, and fifteen deterministic fault/recovery rehearsals passed.

The result is a technical pilot, not a human-subject study. No participant sessions, comprehension outcomes, demand measures, or usability rates were collected. Lifecycle times are synthetic scenario clocks and cannot estimate operational latency. The local integration used the then-current reference composition and does not by itself prove a completed canonical protocol-graph execution path. Subsequent engineering work separately addresses canonical module composition, deployment manifests, and full financial E2E evidence.

10 Decision rules and limitations

A future evaluation may conclude:

- **GO**: registered coverage and operational gates pass within predetermined limits;
- **RESTRICTED GO**: only specified markets, sizes, or leverage tiers pass;
- **PAUSE**: instrumentation or sample coverage is insufficient;
- **NO-GO**: the eligible operating region is empty or uneconomic.

No-go is not a failed study. It is the correct conclusion when venue or capital evidence does not support leverage. When a probabilistic admission mode is evaluated, its confidence level, exchangeability unit, and residual α-risk must be reported separately from any deterministic certificate.

The present pilot has major limitations. It uses no live venue writes, real assets, human sessions, public liquidity, or production credit. It does not estimate fill quality, settlement reliability, actual-close behavior, provider elasticity, market-maker delivery, liquidator abundance, reserve sufficiency, default probability, or commercial demand. The AEMB and AVET releases are deterministic synthetic evidence, not substitutes for venue data. Public books also do not automatically confer rights to redistribute raw payloads.

The analysis is additionally vulnerable to selection and dependence. Markets that survive data-quality gates may be atypical; observations cluster by venue, event, market maker, stablecoin, and close window; and execution policies affect the book they measure. Holdout windows, cluster-aware uncertainty, and full breach reporting are therefore essential.

11 Conclusion

The empirical problem for event-margin protocols is not reducible to a single backtest. It requires a chain of evidence: capability, book reconstruction, match and settlement, actual close, aggregate execution, capital supply, liquidation, withdrawals, and operational recovery. This paper supplies that chain as a prospectively frozen design and demonstrates that the instrumentation can preserve deterministic accounting and fail-closed boundaries across a synthetic Internal Alpha.

The pilot is intentionally narrow. It establishes that the measurement system can distinguish attempted from admitted journeys, settlement from match, expected from observed state, and evidence conflict from financial mutation. It does not establish live performance. The next scientific stage is a rights-cleared sandbox or read-only observation programme followed by held-out evaluation. Only then can the protocol estimate whether a non-empty, capital-efficient 2× validation region exists and whether higher tiers should remain shadow-only or be rejected. Such evidence can validate or falsify the policy envelope; it does not convert a marginal coverage statement into a universal debt-clearing theorem.

12 Evidence taxonomy and falsification policy

The study uses four evidence classes that must not be collapsed:

(1) **Formal**: theorem or invariant under stated assumptions;

(2) **Synthetic**: deterministic or stochastic output generated from author-specified scenarios;

(3) **Emulated**: trace produced by the versioned Venue Emulator, possibly from a rights-cleared snapshot;

(4) **Empirical**: observation from a permitted external sandbox or live venue with provenance and accounting-grade evidence.

A result may move from one class to another only by collecting the missing evidence; a stronger label cannot be inferred from agreement with a model.

The practical thesis is weakened if held-out settled proceeds breach the registered envelope too often, actual close routinely precedes hard-flat, aggregate depth is insufficient at commercially relevant size, signer/lien capabilities cannot be obtained, or required provider compensation exceeds trader willingness to pay. A zero eligible set, restricted 2× region, or failed market-maker commitment is publishable evidence rather than a reason to alter the hypotheses.

For the Internal Alpha pilot, falsification is narrower. The release would fail if attempted journeys disappeared from the denominator, actual and expected state hashes diverged without explanation, duplicate or conflicting evidence changed debt, any external write occurred, or production assets moved. Those gates passed for the registered synthetic bundle. They do not answer the live-market questions above.

Data and code availability

AEMB v0.1.0 and AVET v0.2.0 are versioned at https://github.com/AxientLab/aemb/releases/tag/aemb-v0.1.0 and https://github.com/AxientLab/avet/releases/tag/avet-v0.2.0. The sanitised Internal Alpha engineering evidence remains private because it contains operational provenance not intended as a public user-behaviour or live-performance dataset. DOI information will be added in a later revision.

Disclosure

Axient is an active research-and-development initiative of the author. Automated language and code-assistance tools were used for editorial review, consistency checks, LaTeX support, synthetic test generation, and verification. The author specified and reviewed the research design, hypotheses, estimands, evidence boundaries, analysis, and final manuscript, and takes responsibility for the work.

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